WINDOWS SERVER BOSHQARUVI · MUSTAQIL ISH / TAQDIMOT

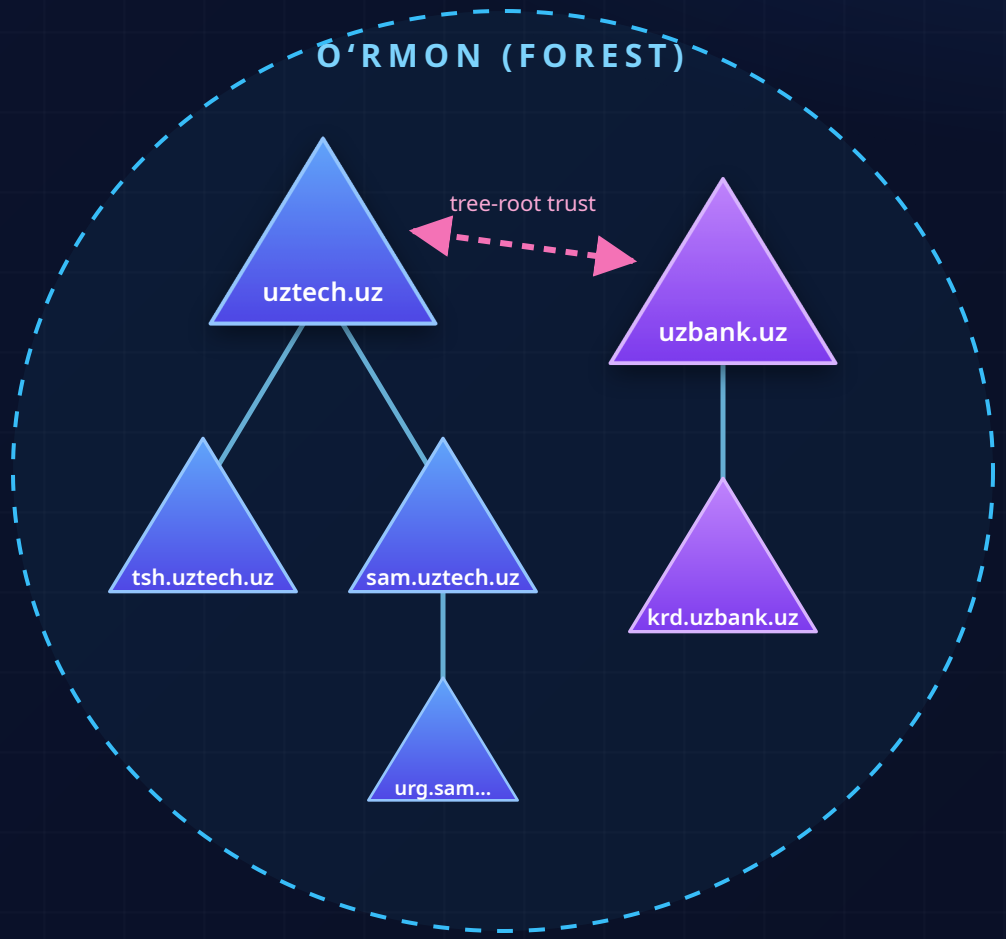
Domen, o‘rmon va daraxt tushunchalari asosida AD tuzilmasini loyihalash

Active Directory Domain Services (AD DS): mantiqiy tuzilma, ishonch munosabatlari, FSMO rollari, OU dizayni va PowerShell orqali amaliy joriy etish

Bajardi
Tolibjon Karimov

Guruh
315-guruh

2026-yil



MUNDARIJA

Taqdimot rejasi

1. Nazariy asoslar

AD DS nima, mantiqiy va fizik tuzilma, ierarxiya

2. Domen · Daraxt · O‘rmon

Nomlash, DC, NTDS.dit, SYSVOL, chegaralar

3. Ishonch va rollar

Trust turlari, FSMO, Global Catalog, sxema, funksional darajalar

4. Loyihalash metodologiyasi

O‘rmon, domen va OU dizayn modellari, saytlar

5. Amaliy keys: «UzTech Group»

Talablar » yechim » OU tuzilmasi » replikatsiya

6. PowerShell va xavfsizlik

Joriy etish skriptlari, tekshirish, Tier modeli, xulosa

Active Directory Domain Services nima?

AD DS — Microsoft tomonidan ishlab chiqilgan **katalog xizmati** bo‘lib, tarmoqdagi obyektlar (foydalanuvchi, kompyuter, guruh, printer) haqidagi ma’lumotni ierarxik bazada saqlaydi va ularga markazlashgan holda **autentifikatsiya** hamda **avtorizatsiya**ni ta’minlaydi.

- Birinchi marta **Windows 2000 Server** bilan taqdim etilgan
- Protokollar: `LDAP` (389/636), `Kerberos v5` (88), `DNS` (53), `SMB` (445), `RPC` (135)
- Standart: X.500 katalog modeliga asoslangan
- Joylashuv xizmati sifatida **DNS**ga to‘liq tayanadi (SRV yozuvlar)
- Guruh siyosati (GPO) orqali minglab kompyuterni markaziy boshqarish



AD tuzilmasi: mantiqiy va fizik qatlam

Mantiqiy tuzilma — «nima va kim»

Tashkilotning ma’muriy modeli. Tarmoq topologiyasiga bog‘liq emas.

Komponent	Vazifasi
O‘rmon (Forest)	Eng yuqori konteyner, xavfsizlik chegarasi
Daraxt (Tree)	Uzluksiz DNS nomlar fazosiga ega domenlar
Domen (Domain)	Ma’muriy va replikatsiya birligi
OU	Vakolat berish va GPO qo‘llash konteyneri
Obyektlar	User, Computer, Group, Contact…

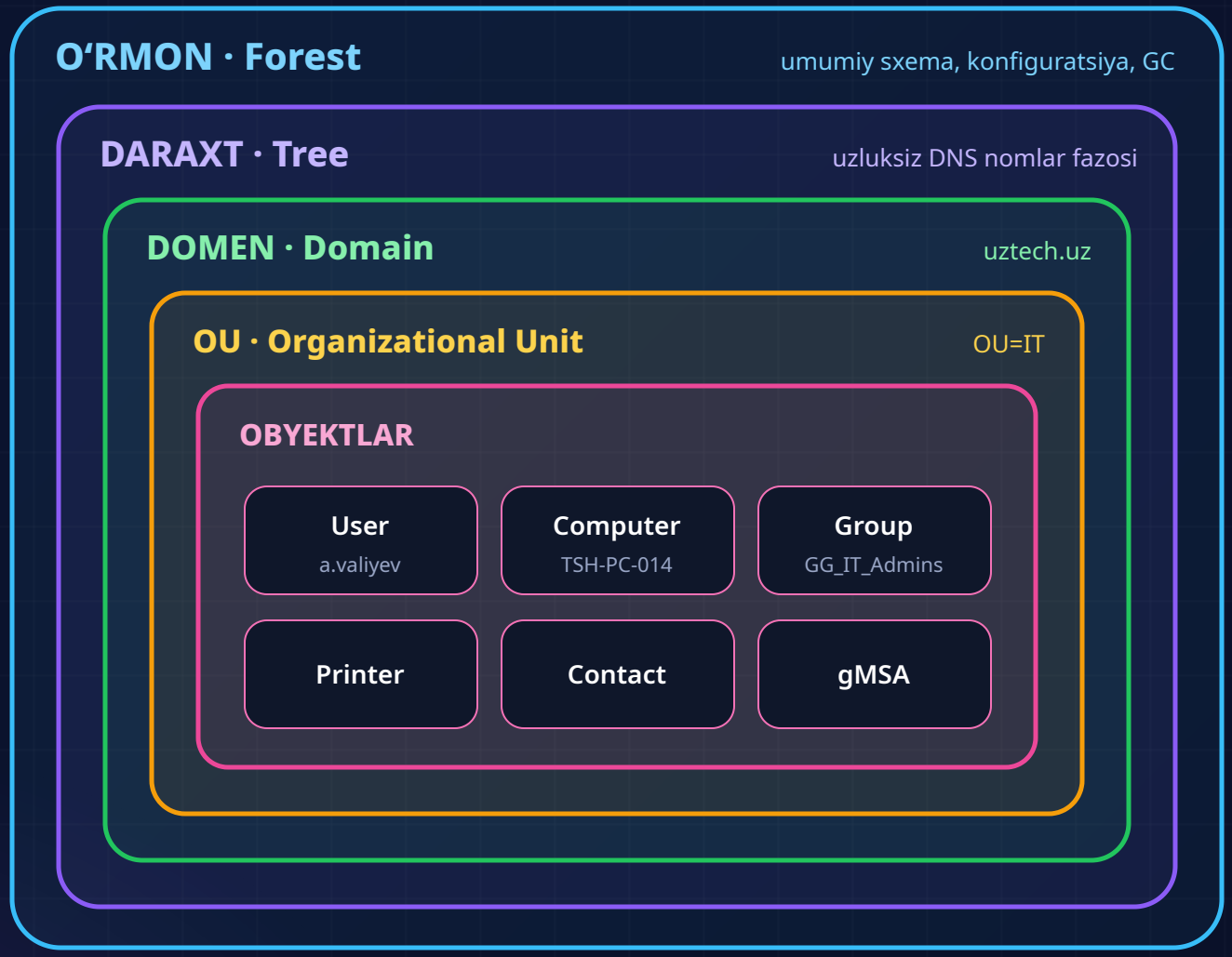
Fizik tuzilma — «qayerda va qanday»

Tarmoq infratuzilmasi: replikatsiya va kirish trafigini boshqaradi.

Komponent	Vazifasi
Domain Controller	AD bazasi nusxasini saqlovchi server
Sayt (Site)	Tez ulangan IP-subnetlar to‘plami
Subnet	Klientni eng yaqin DC ga yo‘naltiradi
Site Link	Saytlararo replikatsiya yo‘li va narxi
Bridgehead	Saytlararo replikatsiya serveri

Muhim: bitta domen bir nechta saytda, bitta saytda esa bir nechta domen DC lari bo‘lishi mumkin — ikki qatlam bir-biridan mustaqil loyihalanadi.

Mantiqiy ierarxiya: o‘rmondan obyektgacha



Ichma-ich joylashuv

Har bir daraja yuqoridagisining ichida joylashadi. Obyekt har doim aniq **bitta** domenga tegishli.

Meros

GPO va ruxsatlar (ACL) yuqoridan pastga meros bo‘lib o‘tadi: Site » Domain » OU.

Distinguished Name

`CN=Ali Valiyev,OU=IT,DC=uztech,DC=uz`

Domen (Domain) — AD ning asosiy birligi

Domen — umumiy katalog bazasi, xavfsizlik siyosati va ishonch munosabatlariga ega obyektlarning **ma’muriy guruhi**. Har bir domen DNS domen nomiga mos keladi.

1

Ma’muriy chegara

Domain Admins faqat o‘z domenini boshqaradi

2

Replikatsiya birligi

Domain partition faqat shu domen DC lariga tarqaladi

3

Parol siyosati

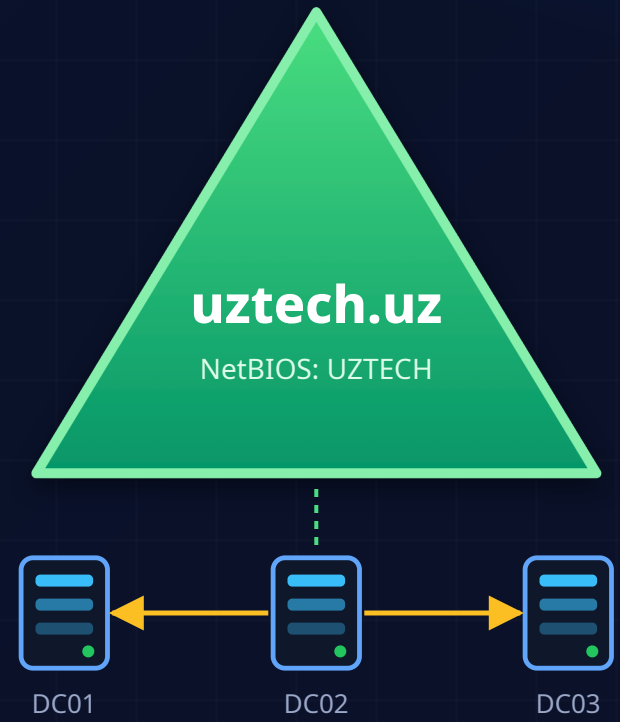
Default Domain Policy; nozik sozlash uchun PSO (FGPP)

4

Autentifikatsiya

Kerberos KDC har bir DC da ishlaydi

> Microsoft tavsiyasi: **xavfsizlik chegarasi — o‘rmon**, domen emas. Bir domendagi Domain Admin butun o‘rmonga ta’sir o‘tkaza oladi.



multi-master replikatsiya

Domen va obyektlarni nomlash qoidalari

Nom turi	Misol
DNS (FQDN)	`sam.uztech.uz`
NetBIOS (<=15 belgi)	`SAMUZTECH`
Distinguished Name	`CN=PC01,OU=Kompyuterlar,DC=sam,DC=uztech,DC=uz`
UPN	`a.valiyev@uztech.uz`
sAMAccountName	`SAM\a.valiyev`
Canonical Name	`sam.uztech.uz/Kompyuterlar/PC01`

> Tavsiya: ro‘yxatdan o‘tgan ommaviy domenning **subdomeni**ni ishlating (masalan `ad.uztech.uz` yoki `corp.uztech.uz`). `.local` dan qoching — u mDNS bilan to‘qnashadi va ommaviy SSL sertifikat olib bo‘lmaydi.

DNS SRV yozuvlari — klient DC ni shu orqali topadi

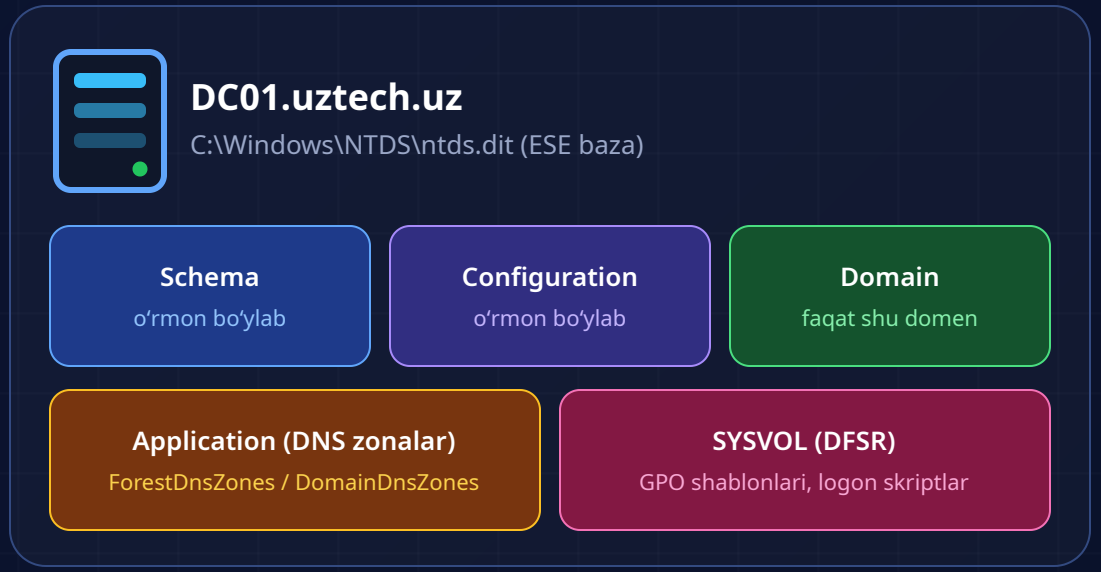
```
; _msdcs zonasi (avtomatik yaratiladi)
_ldap._tcp.dc._msdcs.uztech.uz       SRV 0 100 389  dc01.uztech.uz.
_kerberos._tcp.dc._msdcs.uztech.uz   SRV 0 100 88   dc01.uztech.uz.
_gc._tcp.uztech.uz                   SRV 0 100 3268 dc01.uztech.uz.
_ldap._tcp.Toshkent._sites.uztech.uz SRV 0 100 389  dc01.uztech.uz.
```

PowerShell — tekshirish

```
# DC ni DNS orqali topish
Resolve-DnsName -Type SRV _ldap._tcp.dc._msdcs.uztech.uz
nltest /dsgetdc:uztech.uz

# Obyektning DN ini olish
Get-ADUser a.valiyev | Select DistinguishedName, UserPrincipalName
```

Domain Controller va AD ma’lumotlar bazasi



AD bazasi **mantiqiy bo‘limlar** (naming context / partition) ga ajratilgan; har biri o‘z doirasida replikatsiya qilinadi.

DC dagi asosiy fayllar

```
# Baza va tranzaksiya jurnallari
C:\Windows\NTDS\ntds.dit      # asosiy baza
C:\Windows\NTDS\edb.log       # joriy jurnal
C:\Windows\NTDS\edb.chk       # checkpoint
C:\Windows\SYSVOL\domain\Policies

# Bo‘limlarni ko‘rish
Get-ADRootDSE | Select -Expand namingContexts
```

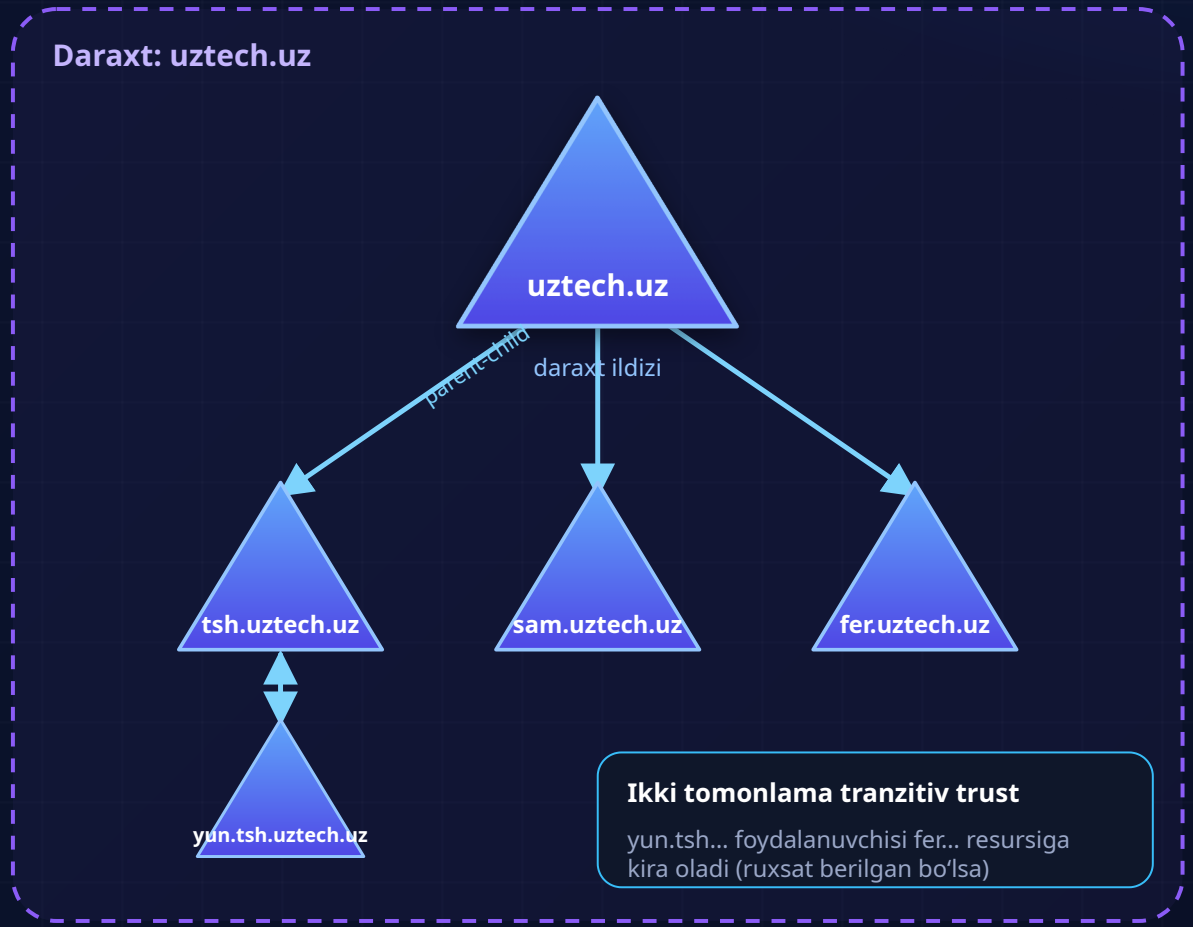
Asosiy faktlar

- **Multi-master**: har bir yozuvchi DC o‘zgarishni qabul qiladi (USN + versiya)
- **RODC** — faqat o‘qish uchun DC, filiallar uchun
- Tombstone lifetime: **180 kun** (yangi o‘rmonlarda)
- WS 2025: yangi o‘rnatishlarda **32k sahifali** baza formati

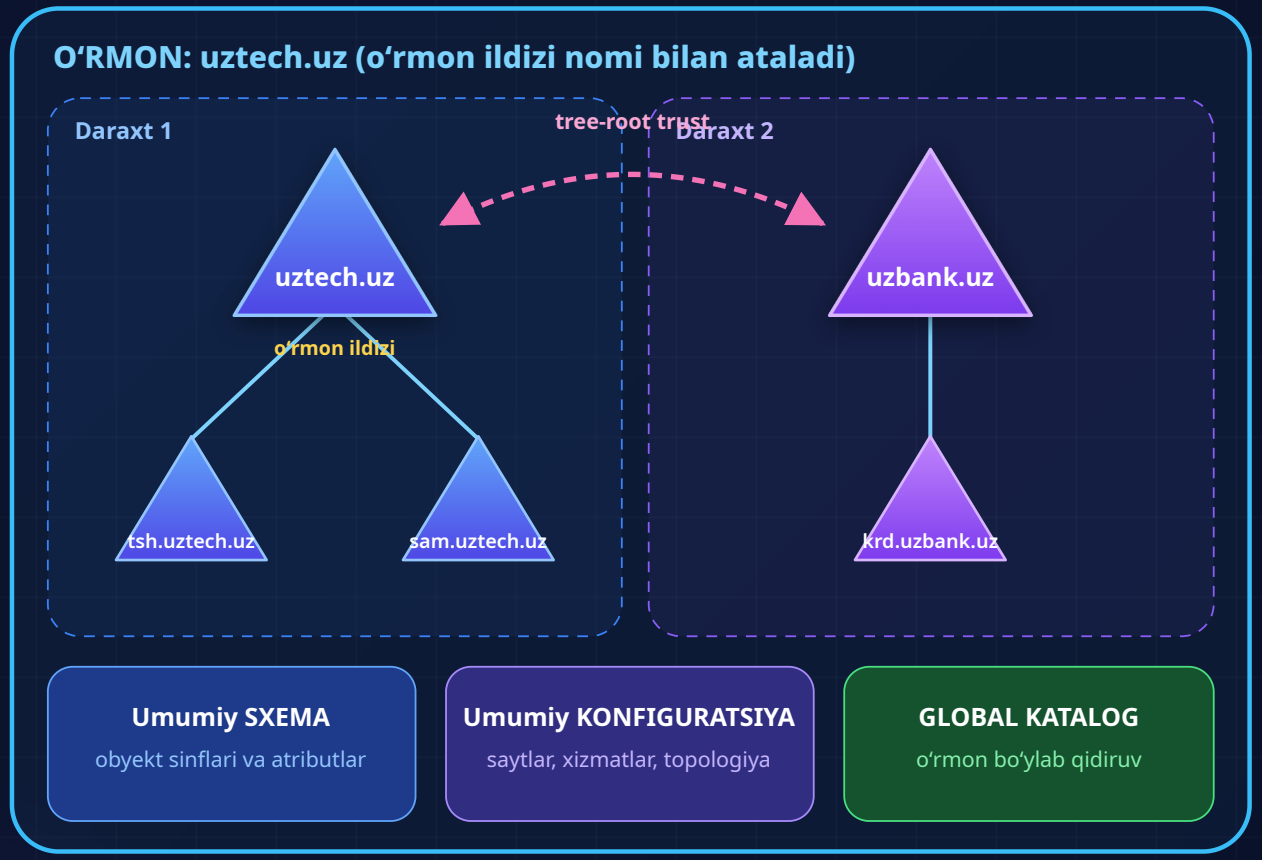
Daraxt (Domain Tree) — uzluksiz nomlar fazosi

Daraxt — umumiy, **uzluksiz (contiguous) DNS nomlar fazosi**ga ega bo‘lgan, ota-bola (parent–child) munosabati bilan bog‘langan domenlar to‘plami.

- Bola domen nomi ota domen nomini o‘z ichiga oladi: `sam.uztech.uz` » `uztech.uz`
- Ota va bola orasida avtomatik **ikki tomonlama tranzitiv** trust yaratiladi
- Daraxtning birinchi domeni — **daraxt ildizi** (tree root)
- Barcha domenlar o‘rmonning umumiy sxema va konfiguratsiyasidan foydalanadi
- Qachon kerak: filiallar alohida ma’muriyat yoki replikatsiya izolyatsiyasini talab qilganda



O‘rmon (Forest) — AD ning eng yuqori konteyneri



O‘rmon — bir yoki bir nechta daraxtdan iborat, ammo **uzluksiz bo‘lmagan** (disjoint) nomlar fazolariga ega bo‘lishi mumkin bo‘lgan tuzilma.

Umumiy jihatlar

- Bitta sxema
- Bitta konfiguratsiya bo‘limi
- Global Catalog
- Barcha domenlar orasida tranzitiv trust

O‘rmon = xavfsizlik chegarasi

To‘liq izolyatsiya faqat alohida o‘rmon bilan ta’minlanadi.

O‘rmon ildiz domeni va maxsus guruhlar

O‘rmonda yaratilgan **birinchi domen** — o‘rmon ildiz domeni. Uni o‘rmonni to‘liq qayta qurmasdan **o‘chirib yoki almashtirib bo‘lmaydi**, shu sababli nomi puxta tanlanishi kerak.

Guruh	Qayerda	Vakolati
Enterprise Admins	faqat ildizda	O‘rmondagi barcha domenlar ustidan to‘liq nazorat
Schema Admins	faqat ildizda	Sxemani o‘zgartirish huquqi
Domain Admins	har bir domenda	O‘z domeni ustidan to‘liq nazorat
Administrators	har bir domenda	DC lar ustidan mahalliy nazorat
Protected Users	har bir domenda	NTLM, DES/RC4, delegatsiyani bloklaydi

Ajratilgan ildiz (Dedicated Forest Root)

Ildiz domenida foydalanuvchilar saqlanmaydi; u faqat o‘rmon darajasidagi guruhlar va rollar uchun xizmat qiladi.

+ izolyatsiya
+ kichik hajm
+ barqaror nom
– qo‘shimcha DC
– murakkablik

O‘rmon darajasidagi guruhlarni tekshirish

```
# Schema Admins bo‘sh bo‘lishi kerak (zarurat bo‘lganda qo‘shiladi)
Get-ADGroupMember "Schema Admins"
Get-ADGroupMember "Enterprise Admins" -Recursive |
  Select Name, SamAccountName

# O‘rmon ildizini aniqlash
(Get-ADForest).RootDomain
```

Domen, daraxt va o‘rmon: taqqoslash

Mezon	Domen	Daraxt	O‘rmon
Ta’rif	Umumiy baza va siyosatli obyektlar guruhi	Uzluksiz DNS nomli domenlar ierarxiyasi	Bir yoki bir nechta daraxt to‘plami
Nomlar fazosi	Bitta DNS nomi	Uzluksiz (contiguous)	Uzluksiz bo‘lmasligi mumkin (disjoint)
Chegara turi	Ma’muriy, replikatsiya, parol siyosati	Nomlash chegarasi	**Xavfsizlik** chegarasi
Sxema	O‘rmonnikini ishlatadi	O‘rmonnikini ishlatadi	Yagona, umumiy
Trust	—	Parent-child (avtomatik)	Tree-root (avtomatik); boshqa o‘rmon bilan forest trust
Admin guruh	Domain Admins	—	Enterprise / Schema Admins
FSMO	RID, PDC, Infrastructure	—	Schema, Domain Naming
Misol	`sam.uztech.uz`	`uztech.uz` + bolalari	`uztech.uz` + `uzbank.uz`

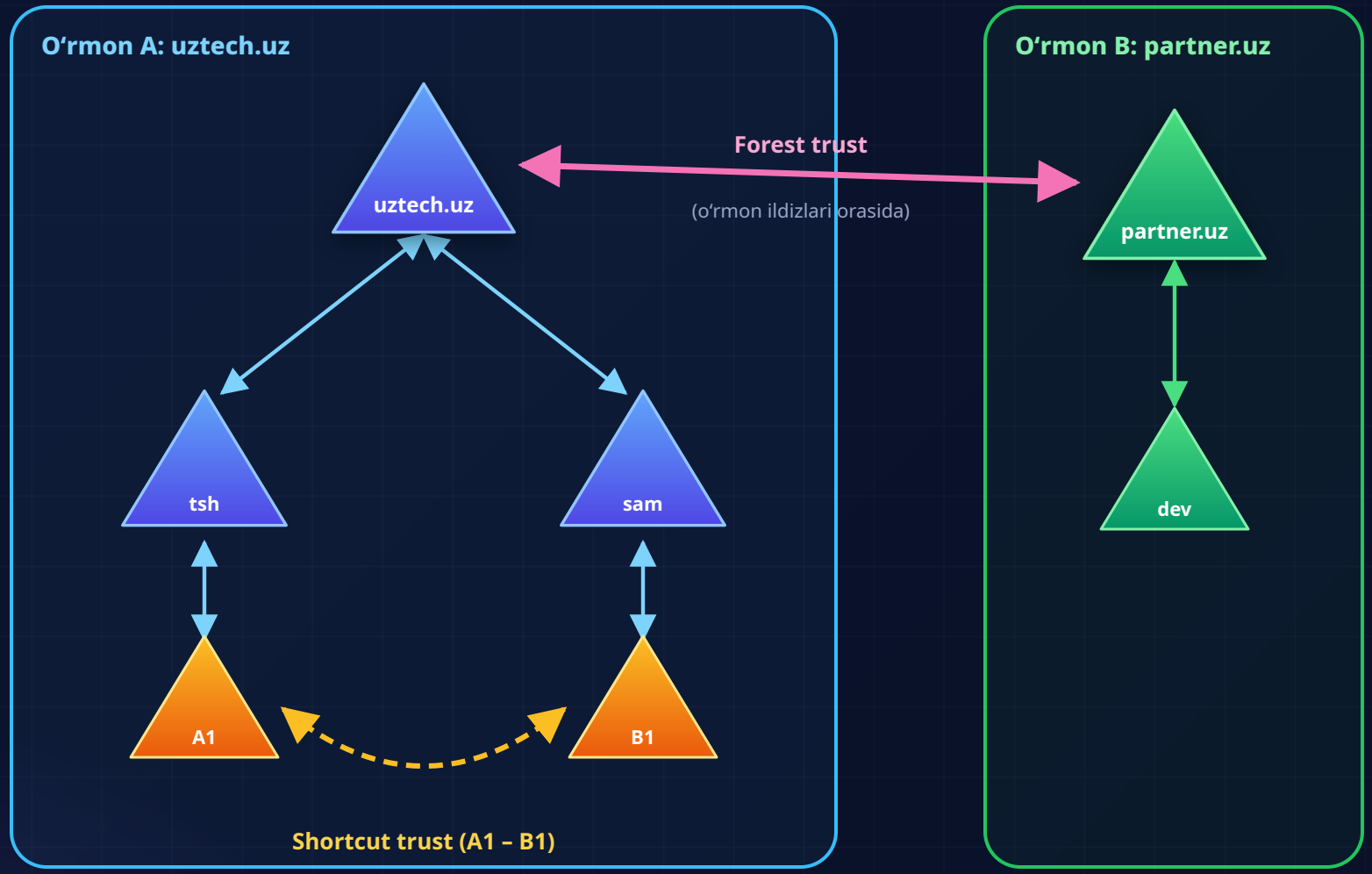
Ishonch (Trust) turlari

Trust — bir domen (ishonuvchi, *trusting*) boshqa domen (ishonchli, *trusted*) foydalanuvchilarining autentifikatsiyasini qabul qilishi. Ruxsat (access) alohida beriladi.

Turi	Yo‘nalish	Tranzitivlik	Yaratilishi	Qo‘llanilishi
Parent-child	Ikki tomonlama	Tranzitiv	Avtomatik	Daraxt ichida ota va bola domen
Tree-root	Ikki tomonlama	Tranzitiv	Avtomatik	Yangi daraxt ildizi – o‘rmon ildizi
Shortcut	Bir / ikki	Tranzitiv	Qo‘lda	O‘rmon ichida autentifikatsiya yo‘lini qisqartirish
Forest	Bir / ikki	Qisman	Qo‘lda	Ikki o‘rmon ildizlari orasida; 3-o‘rmonga o‘tmaydi
External	Bir / ikki	Notranzitiv	Qo‘lda	Boshqa o‘rmondagi aniq bitta domen bilan
Realm	Bir / ikki	Tanlanadi	Qo‘lda	Windows bo‘lmagan Kerberos v5 (MIT, Linux)

Yo‘nalish qoidasi: **kirish (access) yo‘nalishi trust yo‘nalishiga teskari.** A » B ga ishonsa (A trusting), B foydalanuvchilari A resurslariga kira oladi.

Tranzitivlik, shortcut va forest trust



Shortcut bo‘lmasa

A1 » B1 autentifikatsiyasi yo‘li:
A1 » tsh » uztech » sam » B1
4 ta Kerberos referral

Shortcut bilan

A1 » B1 — bevosita, tez, WAN yuklamasi kam

Forest trust

Ikkala o‘rmonning barcha domenlari orasida ishlaydi. Talab: FFL >= 2003, DNS conditional forwarder.

FSMO (Operations Master) rollari

Rol	Doira	Vazifasi
Schema Master	O‘rmon · 1 ta	Sxemaga yozish huquqiga ega yagona DC
Domain Naming Master	O‘rmon · 1 ta	Domen qo‘shish/o‘chirish, application partitionlar
RID Master	Har domen	DC larga RID pul (standart 500 ta) ajratadi » SID yaratish
PDC Emulator	Har domen	Vaqt manbai (o‘rmon ildizidagisi — asosiy), parol o‘zgarishlari, account lockout, GPO tahriri
Infrastructure Master	Har domen	Domenlararo obyekt havolalarini (phantom) yangilaydi

> Formula: o‘rmondagi FSMO rollar soni = **2 + 3 × (domenlar soni)**. 3 domenli o‘rmonda — 11 ta rol.

Rollarni ko‘rish va ko‘chirish

```
# Barcha rollar egasini ko‘rish
netdom query fsmo

Get-ADForest | Select SchemaMaster,
  DomainNamingMaster
Get-ADDomain | Select PDCEmulator,
  RIDMaster, InfrastructureMaster

# Rolni boshqa DC ga ko‘chirish (transfer)
Move-ADDirectoryServerOperationMasterRole `
  -Identity "DC02" `
  -OperationMasterRole PDCEmulator, RIDMaster

# DC ishdan chiqqanda — tortib olish (seize)
Move-ADDirectoryServerOperationMasterRole `
  -Identity "DC02" -OperationMasterRole 0,1,2,3,4 -Force
```

Global Catalog va Sxema

Global Catalog (GC)

- O‘rmondagi **barcha** obyektlarning qisman (PAS) nusxasi
- Port: `3268` (LDAP), `3269` (LDAPS)
- Universal guruh a’zoligini tekshiradi » logon uchun zarur
- UPN orqali kirishda domenni aniqlaydi
- Har bir saytda kamida bitta GC tavsiya etiladi (yoki **Universal Group Membership Caching**)



Sxema (Schema)

Obyekt **sinflari** (classSchema) va **atributlari** (attributeSchema) ta’rifi. O‘rmon bo‘ylab bitta va faqat **kengaytiriladi** — sinf/atributni o‘chirib bo‘lmaydi, faqat «defunct» qilinadi.

Sxema versiyasini aniqlash

```
Get-ADObject (Get-ADRootDSE).schemaNamingContext `
  -Property objectVersion | Select objectVersion

# 87 = WS2016 · 88 = WS2019/2022 · 91 = WS2025
# Sxemani tayyorlash (yangilash oldidan)
adprep /forestprep
adprep /domainprep
```

Domen va o‘rmon funksional darajalari (DFL / FFL)



- **DFL** — domendagi DC lar minimal OS versiyasini va domen imkoniyatlarini belgilaydi
- **FFL** — o‘rmondagi barcha domenlar uchun; DFL hech qachon FFL dan past bo‘lmaydi
- WS 2019 va WS 2022 uchun **alohida daraja yo‘q** — ular 2016 darajasida ishlaydi
- WS 2025 DC qo‘shish uchun FFL kamida **Windows Server 2016** bo‘lishi kerak
- Darajani ko‘tarish — bir tomonlama amal; odatda orqaga qaytarib bo‘lmaydi

Darajani tekshirish va ko‘tarish

```
# Joriy darajalar
(Get-ADForest).ForestMode
(Get-ADDomain).DomainMode

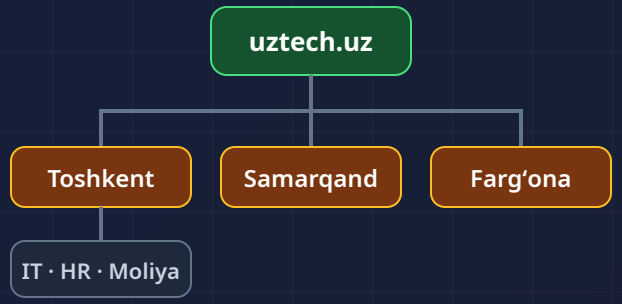
# Avval domen, keyin o‘rmon
Set-ADDomainMode -Identity uztech.uz `
  -DomainMode Windows2025Domain

Set-ADForestMode -Identity uztech.uz `
  -ForestMode Windows2025Forest

# AD Recycle Bin ni yoqish
Enable-ADOptionalFeature 'Recycle Bin Feature' `
  -Scope ForestOrConfigurationSet `
  -Target uztech.uz
```

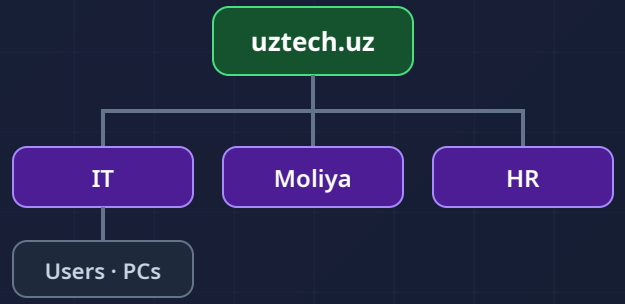
Organizational Unit (OU) dizayn modellari

Geografik model



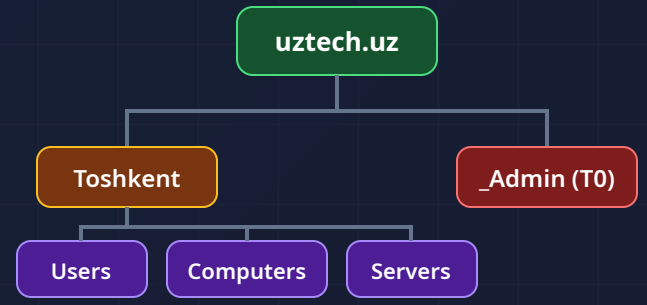
Filiallarda mahalliy adminlar bo‘lsa qulay. Tuzilma barqaror.

Funksional model



Bo‘limlarga xos GPO lar uchun qulay; qayta tashkil etishda o‘zgaradi.

Gibrid model tavsiya



Joylashuv » obyekt turi. Delegatsiya va GPO ni aniq ajratadi.

OU ning ikki asosiy maqsadi: **vakolatni topshirish** (Delegation of Control) va **GPO bog‘lash**. Chuqurlik 3–5 darajadan oshmagani ma’qul; standart `CN=Users` va `CN=Computers` konteynerlariga GPO bog‘lab bo‘lmaydi.

AD mantiqiy tuzilmasini loyihalash bosqichlari



Biznes talablar

- Tashkiliy tuzilma, filiallar
- Qonunchilik (ma’lumot izolyatsiyasi)
- Qo‘shilish/ajralish rejalari

Texnik talablar

- WAN tezligi va ishonchliligi
- Foydalanuvchi/kompyuter soni
- Mavjud ilovalar (sxema talabi)

Ma’muriy talablar

- Markaziy yoki taqsimlangan IT
- Avtonomiya vs izolyatsiya
- Vakolatlar matritsasi

Natija (hujjat)

- O‘rmon/domen diagrammasi
- Nomlash standarti
- OU va GPO xaritasi

O‘rmon dizayn modellari

Tashkiliy o‘rmon

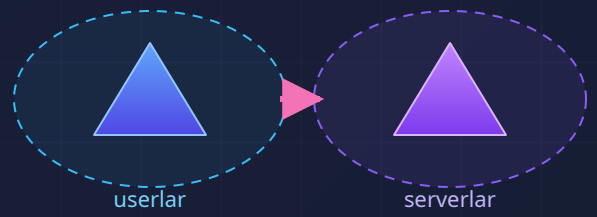
Organizational forest model



Foydalanuvchilar ham, resurslar ham bitta o‘rmonda. **Eng keng tarqalgan**; avtonomiya beradi.

Resurs o‘rmoni

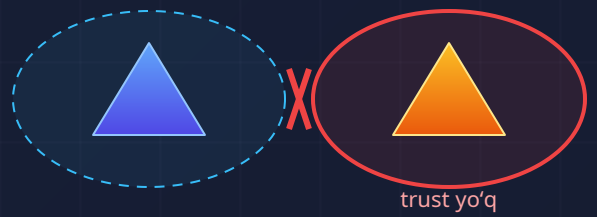
Resource forest model



Resurslar alohida o‘rmonda, foydalanuvchilar trust orqali kiradi. Muhim xizmatlarni ajratish uchun.

Cheklangan kirish o‘rmoni

Restricted access forest



To‘liq **izolyatsiya**: trust yo‘q. Davlat siri, bank protsessing kabi tizimlar uchun.

Bitta o‘rmon tanlang, agar: yagona IT boshqaruv, umumiy sxema talabi, oddiy boshqaruv kerak.

Ko‘p o‘rmon tanlang, agar: huquqiy izolyatsiya, mustaqil sxema yoki ishonchsiz adminlar mavjud.

Domen dizayn modellari: bitta yoki bir nechta?

Mezon	Bitta domen	Mintaqaviy domenlar
Boshqaruv	Oddiy	Murakkab
DC soni	Kam	Ko‘p (har domenga >=2)
Replikatsiya	Barcha DC larga butun domen	Domen ma’lumoti mintaqada qoladi
Parol siyosati	Bitta (+FGPP)	Har domenda alohida
Obyektlarni ko‘chirish	OU orasida oson	ADMT talab qiladi
Mos keladi	Ko‘pchilik tashkilotlar	Sekin WAN, katta hajm, til/qonun farqi

> Microsoft tavsiyasi: **imkon qadar bitta domen**. Domen qo‘shish faqat aniq texnik yoki huquqiy sabab bo‘lganda oqlanadi.

Qaror daraxti



Keys: «UzTech Group» xoldingi — talablar

3 200
foydalanuvchi

3
shahar (filial)

2
yuridik shaxs

Joylashuv	Foydalanuvchi	WAN	IT xodimi
Toshkent (bosh ofis)	2 000	—	Markaziy IT
Samarqand	800	100 Mbit/s	Mahalliy admin
Farg‘ona	250	20 Mbit/s	Yo‘q
UzBank (sho‘ba bank)	150	Toshkentda	Alohida IT

T1 · Yagona tizimga kirish

Barcha xodimlar bitta login bilan umumiy portal va pochtaga kirsin

T2 · Bank boshqaruvi mustaqil

Bank adminlari o‘z domenini boshqarsin, alohida DNS nomi: `uzbank.uz`

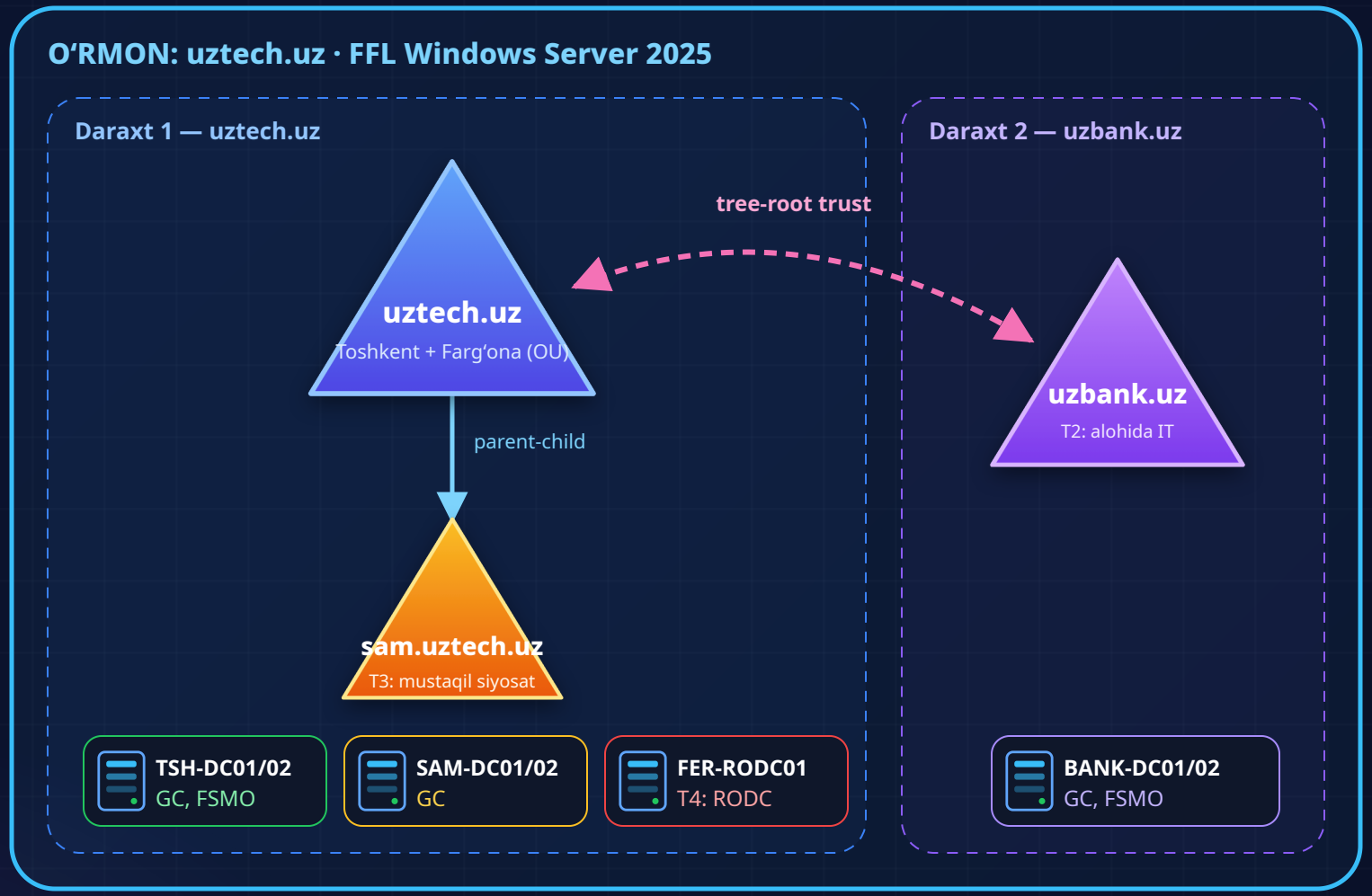
T3 · Samarqand avtonomiyasi

Mahalliy admin faqat o‘z filialini boshqarsin, alohida parol siyosati

T4 · Farg‘onada xavfsizlik

Server jismoniy himoyasiz, IT xodimi yo‘q

Yechim: bitta o‘rmon, ikki daraxt, uchta domen



Nega bitta o‘rmon?

T1 — umumiy GC va tranzitiv trust orqali yagona kirish. To‘liq izolyatsiya talab qilinmagan.

Nega yangi daraxt?

T2 — bank boshqa DNS nomiga (`uzbank.uz`) muhtoj; domen chegarasi alohida Domain Admins beradi.

Nega bola domen?

T3 — alohida ma’muriyat va domen darajasidagi siyosat.

Nega Farg‘ona — OU + RODC?

T4 — 250 kishi uchun domen ortiqcha; RODC parollarni keshlamaydi (PRP bilan cheklanadi).

uztech.uz domenining OU tuzilmasi

Active Directory Users and Computers — uztech.uz

```
uztech.uz
+-- _Admin                # Tier 0 — faqat Domain Admins
|   +-- Accounts          # adm-* imtiyozli hisoblar
|   +-- Groups
|   \-- PAW               # Privileged Access Workstations
+-- Toshkent
|   +-- Users : IT · HR · Moliya · Savdo
|   +-- Computers : Desktops · Laptops
|   +-- Servers
|   \-- Groups
+-- Fargona               # RODC filiali
|   +-- Users
|   \-- Computers
+-- Service Accounts      # gMSA / dMSA
\-- Disabled              # 90 kunlik saqlash
```

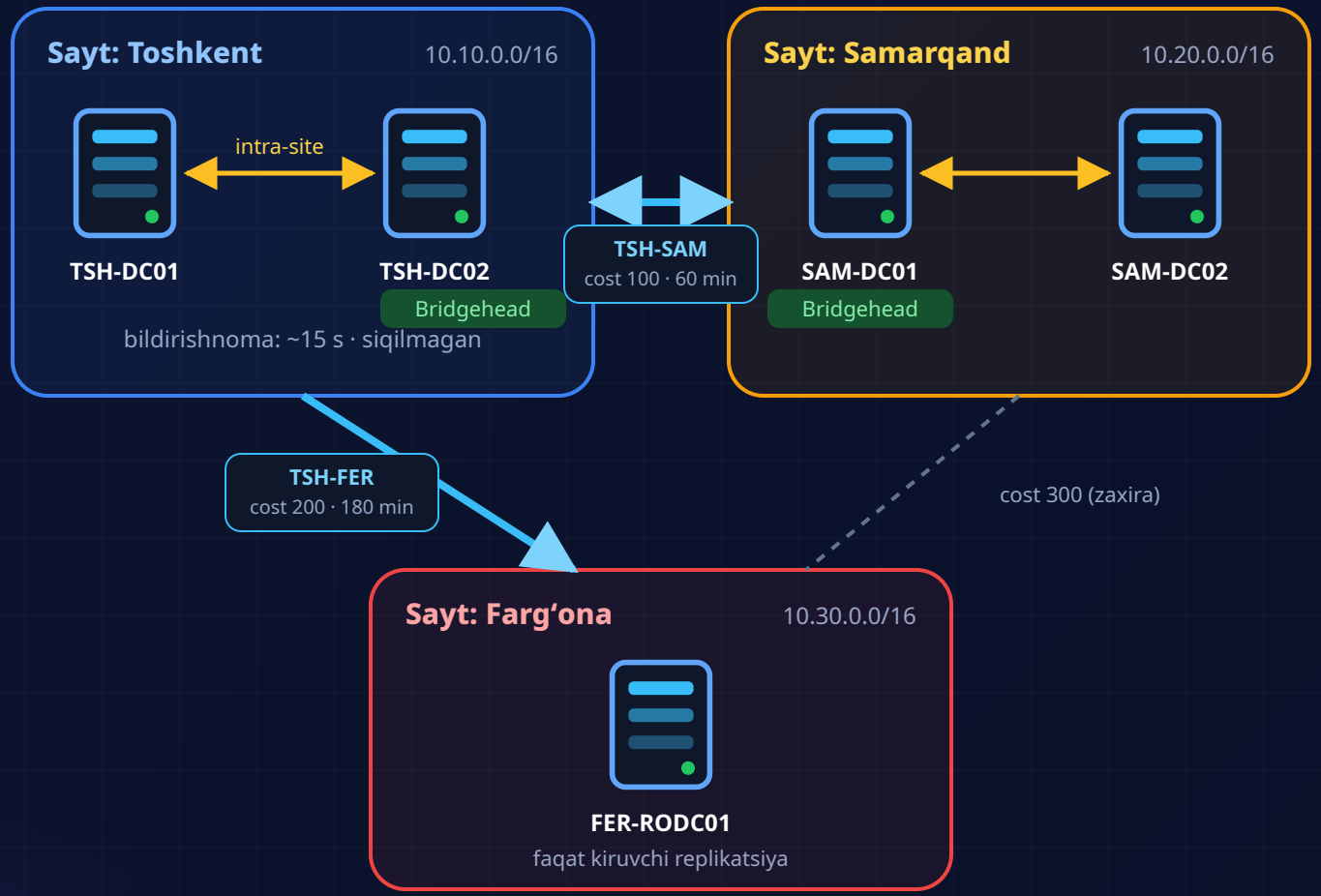
OU	Bog‘langan GPO
uztech.uz	Default Domain Policy, Parol: 14 belgi
_Admin	T0 — Deny logon to T1/T2
***/Computers**	BitLocker, Defender, LAPS
***/Servers**	Server Baseline, Audit
***/Users**	Drive mapping, Desktop

Guruhlarni nomlash (AGDLP)

Account » **G**lobal » **D**omain **L**ocal » **P**ermission

`GG_TSH_Moliya` » `DL_FS01_Moliya_RW` » NTFS

Saytlar, subnetlar va replikatsiya topologiyasi



KCC

Knowledge Consistency Checker har 15 daqiqada replikatsiya topologiyasini avtomatik quradi; ISTG saytlararo ulanishlarni boshqaradi.

Site Link Cost

Standart narx 100, interval 180 daqiqa (minimal 15). Kichik narx = ustuvor yo‘l.

```
New-ADReplicationSite Samarqand
New-ADReplicationSubnet -Name 10.20.0.0/16 `
  -Site Samarqand
New-ADReplicationSiteLink TSH-SAM `
  -SitesIncluded Toshkent,Samarqand `
  -Cost 100 -ReplicationFrequencyInMinutes 60
```

1-qadam: AD DS rolini o‘rnatish va o‘rmon yaratish

TSH-DC01 · Administrator: Windows PowerShell

```
# 0. Statik IP va server nomi (oldindan)
New-NetIPAddress -InterfaceAlias "Ethernet" -IPAddress 10.10.0.10 `
  -PrefixLength 16 -DefaultGateway 10.10.0.1
Rename-Computer -NewName "TSH-DC01" -Restart

# 1. AD DS rolini o‘rnatish
Install-WindowsFeature -Name AD-Domain-Services -IncludeManagementTools

# 2. Yangi o‘rmon yaratish (eski DC lar bo‘lsa: "WinThreshold" = 2016)
Import-Module ADDSDeployment
Install-ADDSForest `
  -DomainName        "uztech.uz" `
  -DomainNetbiosName  "UZTECH" `
  -ForestMode        "Win2025" `
  -DomainMode        "Win2025" `
  -InstallDns:$true `
  -DatabasePath      "C:\Windows\NTDS" `
  -LogPath           "C:\Windows\NTDS" `
  -SysvolPath        "C:\Windows\SYSVOL" `
  -SafeModeAdministratorPassword (Read-Host -AsSecureString "DSRM parol") `
  -Force
```

Natija

- Yangi o‘rmon **uztech.uz** va uning ildiz domeni
- DNS zona va `_msdcs` AD-integratsiyalangan
- DC01: GC + barcha **5 FSMO** roli
- Server avtomatik qayta yuklanadi

DSRM paroli

Directory Services Restore Mode — bazani tiklash uchun. Uni xavfsiz joyda (vault) saqlang, skript ichida ochiq yozmang.

Sinov rejimi

`Test-ADDSForestInstallation` — o‘zgarishsiz tekshiradi.

2-qadam: bola domen, yangi daraxt va qo‘shimcha DC

SAM-DC01 — bola domen (ChildDomain)

```
Install-WindowsFeature AD-Domain-Services -IncludeManagementTools

Install-ADDSDomain `
   -DomainType          ChildDomain `
   -ParentDomainName    "uztech.uz" `
   -NewDomainName       "sam" `
   -NewDomainNetbiosName "SAMUZTECH" `
   -DomainMode          "Win2025" `
   -SiteName            "Samarqand" `
   -InstallDns -CreateDnsDelegation `
   -Credential (Get-Credential "UZTECH\Administrator") `
   -SafeModeAdministratorPassword (Read-Host -AsSecureString)
# » sam.uztech.uz, parent-child trust avtomatik
```

TSH-DC02 — mavjud domenga qo‘shimcha DC

```
Install-ADDSDomainController -DomainName "uztech.uz" `
   -SiteName "Toshkent" -InstallDns -NoGlobalCatalog:$false `
   -Credential (Get-Credential)
```

BANK-DC01 — yangi daraxt (TreeDomain)

```
# ParentDomainName = o‘rmon ildizi, NewDomainName = to‘liq FQDN
Install-ADDSDomain `
   -DomainType          TreeDomain `
   -ParentDomainName    "uztech.uz" `
   -NewDomainName       "uzbank.uz" `
   -NewDomainNetbiosName "UZBANK" `
   -InstallDns `
   -Credential (Get-Credential "UZTECH\Administrator") `
   -SafeModeAdministratorPassword (Read-Host -AsSecureString)
# » tree-root trust avtomatik yaratiladi
```

FER-RODC01 — faqat o‘qish uchun DC

```
Install-ADDSDomainController -DomainName "uztech.uz" `
   -ReadOnlyReplica -SiteName "Fargona" `
   -AllowPasswordReplicationAccountName "GG_FER_Users" `
   -DelegatedAdministratorAccountName "GG_FER_RODC_Admins" `
   -InstallDns -Credential (Get-Credential)
```

> O‘rmonga domen qo‘shish uchun **Enterprise Admins** a’zosi hisobi va **Domain Naming Master** mavjudligi talab qilinadi.

3-qadam: OU tuzilmasi, foydalanuvchilar va guruhlar

Build-OUStructure.ps1

```
$base = (Get-ADDomain).DistinguishedName     # DC=uztech,DC=uz
$sites = "Toshkent", "Fargona"
$sub   = "Users", "Computers", "Servers", "Groups"

New-ADOrganizationalUnit -Name "_Admin" -Path $base
foreach ($s in $sites) {
  New-ADOrganizationalUnit -Name $s -Path $base `
    -ProtectedFromAccidentalDeletion $true
  foreach ($o in $sub) {
    New-ADOrganizationalUnit -Name $o -Path "OU=$s,$base"
  }
}

# CSV dan ommaviy foydalanuvchi yaratish
Import-Csv .\xodimlar.csv | ForEach-Object {
  New-ADUser -Name "$($_.Ism) $($_.Familiya)" `
    -GivenName $_.Ism -Surname $_.Familiya `
    -SamAccountName $_.Login `
    -UserPrincipalName "$($_.Login)@uztech.uz" `
    -Department $_.Bolim `
    -Path "OU=Users,OU=$($_.Filial),$base" `
    -AccountPassword (ConvertTo-SecureString $_.TempPass -AsPlainText -Force) `
    -ChangePasswordAtLogon $true -Enabled $true
}
```

Guruhlar (AGDLP) va delegatsiya

```
$g = "OU=Groups,OU=Toshkent,$base"
New-ADGroup GG_TSH_Moliya -GroupScope Global `
  -GroupCategory Security -Path $g
New-ADGroup DL_FS01_Moliya_RW -GroupScope DomainLocal `
  -GroupCategory Security -Path $g
Add-ADGroupMember DL_FS01_Moliya_RW -Members GG_TSH_Moliya

# Moliya bo‘limi xodimlarini global guruhga
Get-ADUser -Filter "Department -eq 'Moliya'" `
  -SearchBase "OU=Toshkent,$base" |
 ForEach-Object { Add-ADGroupMember GG_TSH_Moliya -Members $_ }

# Helpdesk ga parol tiklash huquqini berish
dsacls "OU=Users,OU=Toshkent,$base" /I:S `
 /G "UZTECH\GG_Helpdesk:CA;Reset Password;user"
```

`xodimlar.csv` : Ism, Familiya, Login, Bolim, Filial, TempPass. Vaqtinchalik parol birinchi kirishda almashtiriladi; CSV faylni ishdan so‘ng o‘chiring.

4-qadam: forest trust va sog‘liqni tekshirish

partner.uz bilan forest trust

```powershell
# 1) Ikkala tomonda DNS conditional forwarder
Add-DnsServerConditionalForwarderZone -Name "partner.uz" `
  -MasterServers 10.50.0.10 -ReplicationScope Forest

# 2) Ikki tomonlama forest trust (.NET API)
$cred   = Get-Credential "PARTNER\Administrator"
$ctx    = New-Object System.DirectoryServices.ActiveDirectory.DirectoryContext(
          "Forest", "partner.uz", $cred.UserName,
          $cred.GetNetworkCredential().Password)
$remote = [System.DirectoryServices.ActiveDirectory.Forest]::GetForest($ctx)
$local  = [System.DirectoryServices.ActiveDirectory.Forest]::GetCurrentForest()
$local.CreateTrustRelationship($remote, "Bidirectional")

# 3) Trustlarni ko‘rish
Get-ADTrust -Filter * | Select Name, Direction, ForestTransitive
nltest /domain_trusts /all_trusts /v
```

Sog‘liqni tekshirish (health check)

```powershell
# O‘rmon va domen haqida umumiy ma’lumot
Get-ADForest | Select Name, ForestMode, Domains, GlobalCatalogs
Get-ADDomainController -Filter * |
  Select Name, Site, IsGlobalCatalog, IsReadOnly

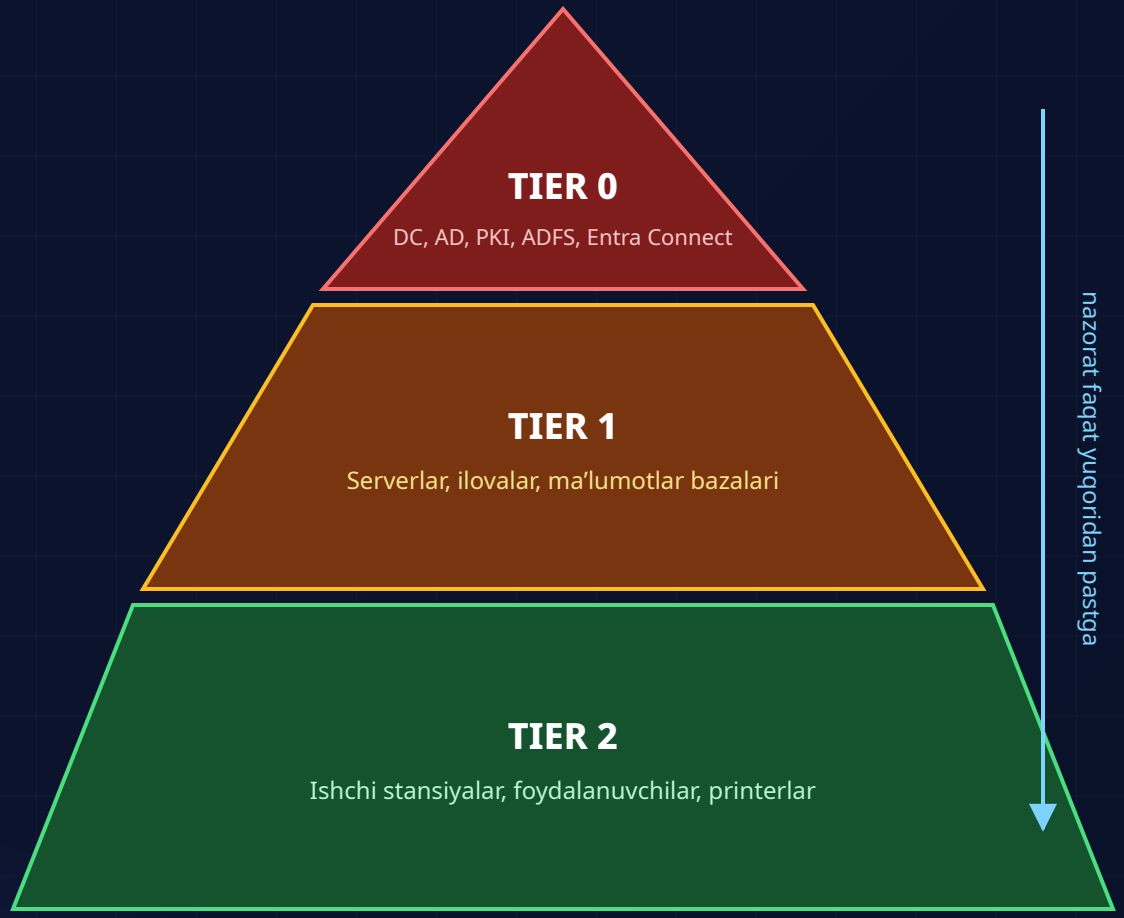
# Replikatsiya holati
repadmin /replsummary
repadmin /showrepl * /csv | ConvertFrom-Csv |
  Where-Object { $_.'Number of Failures' -gt 0 }

# DC diagnostikasi
dcdiag /e /c /q
dcdiag /test:dns /v

# Vaqt sinxronizatsiyasi (Kerberos ±5 daqiqa)
w32tm /monitor
w32tm /query /source
```

> Kerberos uchun soatlar farqi standart **5 daqiqa**dan oshmasligi shart; vaqt iyerarxiyasi PDC Emulator orqali quriladi.

Xavfsiz AD dizayni: Tier modeli (Tier 0/1/2)



Asosiy qoida

Yuqori tier hisobi quyi tier mashinasiga **hech qachon kirmaydi** (credential theft, Pass-the-Hash oldini olish).

Himoya vositalari

- Protected Users
- Windows LAPS
- Authentication Policy Silo
- PAW
- gMSA / dMSA
- LDAP signing
- NTLM ni cheklash

```
# Adminlarni Protected Users ga qo‘shish
Add-ADGroupMember "Protected Users" -Members adm-ali
# Windows LAPS: kompyuter OU siga ruxsat
Set-LapsADComputerSelfPermission `
  -Identity "OU=Computers,OU=Toshkent,DC=uztech,DC=uz"
# Imtiyozli guruhlar auditi
Get-ADGroupMember "Domain Admins" -Recursive
```

Eng yaxshi amaliyotlar va keng tarqalgan xatolar

To‘g‘ri yondashuv

- O‘rmonlar va domenlar sonini **minimal** saqlash
- Har domenda kamida **2 ta DC** (ishonchlilik)
- Ommaviy domenning subdomeni: `ad.uztech.uz`
- Har bir subnetni saytga bog‘lash
- OU ni **ma’muriy model** asosida qurish, org-chart asosida emas
- AD Recycle Bin ni yoqish, System State zaxira nusxasi
- Nomlash standartini hujjatlashtirish
- Virtual DC larda VM-GenerationID dan foydalanish

Xatolar

- `.local` yoki bir qismli nom (`UZTECH`) ishlatish
- Har bir bo‘lim uchun alohida domen yaratish
- Domen chegarasini xavfsizlik chegarasi deb hisoblash
- Infrastructure Master ni GC da qoldirish (ko‘p domenli, hammasi GC bo‘lmasa)
- Schema Admins da doimiy a’zolar
- DC larni snapshotdan tiklash (USN rollback)
- Saytsiz subnetlar » klient uzoqdagi DC ga ulanadi
- DC da boshqa xizmatlar (web, fayl server) ishlatish

XULOSA

Xulosa



Domen

Ma’muriy va replikatsiya birligi. Imkon qadar **bitta** domen + yaxshi o‘ylangan OU.



Daraxt

Uzluksiz DNS nomlar fazosi. Alohida **nom** va brend kerak bo‘lganda yangi daraxt.



O‘rmon

Yagona **xavfsizlik chegarasi**. To‘liq izolyatsiya talab qilinsa — alohida o‘rmon.

AD tuzilmasini loyihalash — **biznes talablaridan** boshlanadi: avval o‘rmon soni, so‘ng domen modeli, DNS nomlari, OU va nihoyat saytlar aniqlanadi. To‘g‘ri dizayn boshqaruv xarajatini kamaytiradi, xavfsizlikni oshiradi va kelajakdagi o‘sishga moslashadi. O‘rmon ildizi va DNS nomini keyinroq o‘zgartirish juda qiyin, shuning uchun bu qarorlar boshida puxta qabul qilinadi.

MANBALAR

Foydalanilgan adabiyotlar

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3. Microsoft Learn. *Active Directory Domain Services Functional Levels.* — …/ad-ds/active-directory-functional-levels
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6. Microsoft Learn. *ActiveDirectory va ADDSDeployment PowerShell modullari.* — learn.microsoft.com/powershell/module/activedirectory
7. [MS-ADTS] *Active Directory Technical Specification.* Microsoft Open Specifications.
8. Desmond B., Richards J., Allen R., Lowe-Norris A. *Active Directory: Designing, Deploying, and Running AD*, 5th ed. — O’Reilly, 2013.
9. Francis D. *Mastering Active Directory*, 3rd ed. — Packt, 2021.
10. Microsoft Learn. *Windows Server 2025 — What’s new in Active Directory Domain Services.*



E’tiboringiz uchun rahmat!

Savollaringiz bo‘lsa, marhamat

Domen Daraxt O‘rmon Trust FSMO OU